

# Harith Al-Safi

Quantitave Software Developer

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## 📄 About me

A Full Stack Engineer with **5 years** of specializing in React.js, Java Spring Boot, Machine Learning and Cloud Technologies. Proven record in leading technical teams, and driving cost optimizations across **startup** to **enterprise**.

## 📁 Experience

### Full stack Engineer

📅 Sep 2024 - Ongoing (16+ Months)

BT Group

📍 Manchester, UK

- Delivered **5 Kafka-based Spring Boot microservices** with MongoDB/PostgreSQL, Netflix Conductor, and **Pulumi**
- Mentored and examined **5+ apprentices** on agile development with **Java 21** Spring Boot and **AWS Lambda**
- Achieved **30% AWS cost reduction** by automating infrastructure with TypeScript and optimizing **ECS cluster**
- Re-architecture Spring Boot APIs with JWT, **Azure AD**, and **role-based access** achieving **enterprise grade security**
- Led **React.js** to **Next.js** migration of 5K+ lines, adding Cypress testing and OpenAPI to TypeScript **code gen**
- Managed **regional Graduate Network** organizing and coordinating monthly tech talks for **30+ attendees**

### Quantitative Trader

📅 Jan 2024 - Ongoing (23+ Months)

ICMarkets FX - Self Managed

📍 Leeds UK

- Analyzed **macroeconomic** (interest rates, inflation, GDP, employment) and quantified effects on **short-term price**.
- Researched **microstructure** using **order flow data**, including footprint charts, volume delta, and **bid/ask imbalance**.
- Worked with **DOM data** to study **liquidity**, **depth dynamics**, and **execution** under varying market conditions.
- Developed **systematic trading strategies** in **Python**, including **risk management**, and **transaction cost modeling**.
- Engineered **automated research and trading pipelines** for **strategy simulation** and **performance evaluation**.

### Robotics Engineer

📅 Dec 2023 - Mar 2024 (4 Months)

Robot Fight League

📍 Leeds, UK

- Implemented a **control system** using Arduino and **C++** for the robot's movement, sensors and pneumatic system
- Tested obstacle detection and tracking using **OpenCV** and Python for the robot's **autonomous navigation** system
- Led **cross-functional** team of 5 engineers to 1<sup>st</sup> place finish, coordinating **hardware-software** integration

### C++ Software Engineer

📅 July 2022 - July 2023 (12 Months)

Johnson Controls

📍 London UK

- Invented a **scripting language with a compiler** for fire detection, increasing development productivity by 40%
- Managed a **team of 10** interns to create a **business plan** and ended up placing 2<sup>nd</sup> amongst the whole company
- Administered C/C++ projects while fixing bugs, adding features, refactoring and **migrating 90% of codebase to C#**
- Developed **C++ and C# libraries** for fire detection simulations and **data analysis** improving operations by 32%
- Collaborated on an **analytics web dashboard with React.js** for predictive maintenance with dev ops and CI/CD
- Maintained Python automation scripts for automated builds of .NET applications and processing data with Azure

### Embedded Systems Machine Learning Engineer

📅 Dec 2021 - May 2022 (6 Months)

Paperound - Freelance

📍 Leeds UK

- Developed **signal processin** filters and modulators using **SciPy Python** and Fourier analysis for RF signals
- Built and trained a reinforcement **neural network** using TensorFlow, **KerasRL** for predicting market movements
- Applied **data visualization** using Python **Matplotlib** to present analytical insights to non-technical stakeholders
- Developed parallel processing system using **C++ OpenMP** to **accelerate polynomial regression** computations
- Developed low level **ARM embedded** systems with **C** whilst using memory allocation and **register programming**

## Education

### BEng (Honours) in Electronics and Computer

📅 Sept 2019 - July 2024

University of Leeds (UoL)

✅ First-Class Honours (4.0/4.0 GPA)

- Cloud and Parallel computing, Secure and Distributed systems, Cybersecurity, Software Engineering, Data Science
- Embedded systems, Microprocessors, FPGA, Circuit Analysis, Signal processing, Networking, Compiler design

## Skills and Languages

**Technical:** C \ C++ Python TypeScript Java C# Bash Matplotlib TensorFlow Scikit-learn  
Numpy SciPy Pandas NLP React.js Next.js Express.js Redux Tailwind CSS Stripe Auth0  
Clerk Spring Boot Kafka .NET Clang OpenMP Cuda Boost QuantLib Docker Jira  
Git MongoDB PostgreSQL Redis SQLite Jenkins Firebase Firestore Pulumi  
AWS (RDS, Lambda, ECS, EC2, S3) Azure (AD, MSAL, Serverless)

**Soft:** Management Leadership Motivated Deterministic Communication Problem solver

**Languages:** Arabic (Full Professional) English (Full Professional) Spanish (Elementary)

## Publications

**Vega: LLM-driven Intelligent Chatbot Platform for Internet of Things Control and Development** 📅 2025

MPDI Sensors - 10.3390/s25123809 - 1<sup>st</sup> Author

- Developed a full stack **Chatbot** web application with TypeScript, React.js, Next.js, **TailwindCSS** and **Reddit**
- Programmed language processing algorithms using **multi-modal large language models** to execute user messages
- Implemented a **Python Flask** REST Server with **SQLite** on the **Raspberry Pi** to communicate with the chatbot
- Engineered a **circuit** with sensors, camera, fans, LCD and buttons to execute user's commands from the chatbot

## Projects

**Green ML: Machine Learning for Green Bond Syndication** 📅 May 2024 - Aug 2024 🔒 Personal

- Engineered bond syndication **prediction model** using **XGBoost** and network analysis on 3K+ institutional holders
- Implemented **parallel processing** with Ray to extract network features from adjacency matrices with **70% speedup**
- Built machine learning pipeline achieving **0.95+ AUC-ROC** for predicting bond syndication collaboration patterns
- Processed **Bloomberg financial data** using **pandas**, **NetworkX** for multi-regional investment analysis

**CV Studio** 📅 Oct 2024 - Ongoing 🔒 Personal - Beta on cvstudio.ai

- Built **full-stack** resume-improvement web app, powered by large language model, serving **hundreds of active users**.
- Implemented **live resume editing** with ShadcnUI and Typst **WebAssembly** PDF compilation for real-time render.
- Integrated **Clerk authentication** and **Stripe subscription** payments for secure user access and monetization
- Designed **Firestore real-time** database with Zustand state and Zod validation for seamless data syncing

**MCP: High Performance C++ Library for Numerical Computation** 📅 Apr 2021 - Oct 2022 🔒 Personal

- Created a **cross-platform math library** for C++ used for matrix algebra, statistics and machine learning.
- Engineered with **CI/CD** using **Jenkins**, unit tests, Docker build, **Cmake** and Doxygen documentation
- Improved performance by 40% using **parallel computing** with **OpenMP**, **OpenCL** and **CUDA**

## Professional Courses

**Data Analysis with Python** 📅 Oct 2021

- Develop Python code for cleaning and preparing data for analysis
- Perform exploratory **data analysis** and analytical techniques to datasets using Pandas, Numpy and Scipy
- Manipulate data using dataframes, understand data distribution, perform correlation and create data pipelines
- Build and evaluate **regression models** using machine learning scikit-learn for prediction and decision making

**Databases and SQL for Data Science with Python** 📅 Sept 2021

- Analyze data within a relational database on the cloud using **IBM DB2**, SQL, Python
- Construct **SQL CRUD** operations and compose views, transactions, stored procedures and joins.

**Python for Data Science, AI & Development** 📅 Aug 2021

- Demonstrate proficiency in using Python libraries such as Pandas, **Numpy**, and Beautiful Soup
- Access web data using APIs and **web scraping** from Python in Jupyter Notebooks.